

Synthetic Data Trade-off

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Cheap data vs. real-world fidelity

PROFESSIONAL DEFINITION

The Synthetic Data Trade-off weighs the low cost and privacy benefits of artificially generated data against its potential loss of fidelity to real-world distributions.

WHO DEVELOPED IT

A construct from machine-learning practice as synthetic data became a common alternative to scarce or sensitive real data.

WHY IT WAS DEVELOPED

It was developed to guide when synthetic data is a sound substitute and when it risks degrading model performance.

FIGURE 1 · SYNTHETIC DATA TRADE-OFF



Figure 1. A schematic of the Synthetic Data Trade-off as applied in BARBM312 (illustrative).

HOW IT IS USED

For a use case, the cost and privacy gains of synthetic data are weighed against fidelity risk and validated against real data where possible.

WHEN IT IS USED

When sourcing training data under cost or privacy constraints.

BY WHOM IT IS USED

ML engineering, data and privacy teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It frames synthetic data as a deliberate trade-off rather than a free substitute, guarding against silent model degradation.

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Data Rights Stack · Slibrary 15

Data Moat Strength · Slibrary 15

Privacy-Value Frontier · Slibrary 15

SOURCES (HARVARD REFERENCING STYLE)

Jordon, J. et al. (2022) 'Synthetic data — what, why and how?', Royal Society / The Alan Turing Institute.

Nikolenko, S.I. (2021) Synthetic Data for Deep Learning. Cham: Springer.